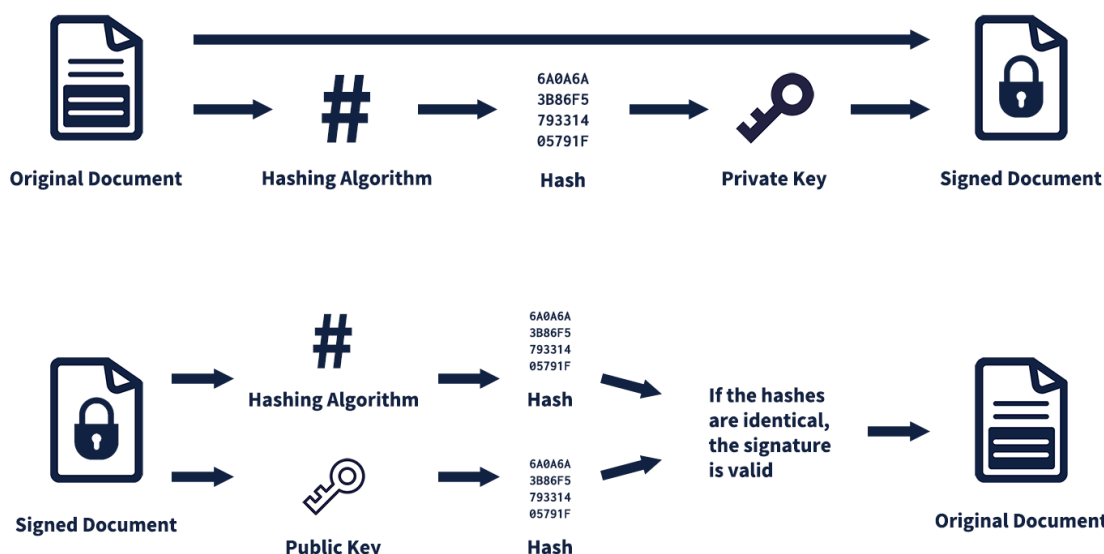


Digital Signature



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A **digital signature** is the ultimate combination of everything we've talked about so far. It brings together **asymmetric cryptography (keys)** and **hash functions** to solve a massive problem online: *How do you prove that a document or message actually came from you, and that no one altered it along the way?*

Do not confuse it with an electronic signature (like typing your name or uploading an image of your cursive signature onto a PDF, as shown in the decorative tablet graphic above). An electronic signature is just an image. A **digital signature** is a mathematical seal of authenticity.

It replaces the ink signature with something far more secure that cannot be forged or copied.

How It Works: A Mathematical Two-Step

A digital signature works by reversing the typical public-key encryption process. Instead of someone using your public key to *lock* a message for you, **you use your private key to sign the message** for the world to verify.

Here is the exact step-by-step process of how Alice digitally signs a contract for Bob:

Step 1: Creation (The Sender's Side)

1. **Hashing:** Alice's computer takes the contract document and runs it through a hash function to get a unique digital fingerprint (a hash value).

2. **Signing:** Alice's computer encrypts that hash value using her **Private Key**. This encrypted hash *is* the digital signature.
3. **Sending:** The signature is attached to the contract, and the whole package is sent to Bob.

Step 2: Verification (The Receiver's Side)

1. **Decrypting the signature:** Bob receives the document and the signature. His computer decrypts the signature using Alice's **Public Key**. This reveals the original hash Alice generated.
2. **Hashing the document:** Bob's computer independently runs the document through the exact same hash function to generate a fresh hash.
3. **The Match:** Bob's computer compares the two hashes.
 - If they match perfectly, Bob knows two things: The document was signed by someone with Alice's private key (Authentication), and not a single character of the contract has been altered since she signed it (Integrity).

The Three Things Digital Signatures Guarantee

Digital signatures are legally binding in most of the world because they provide three ironclad guarantees that ink signatures never could:

- **Authentication:** Because only Alice has her private key, only Alice could have created that exact encrypted hash.
- **Integrity:** Because of the avalanche effect in hashing, if a hacker tries to alter the contract (even changing a price from \$10 to \$100), the hash Bob generates will be entirely different from the decrypted signature hash. The verification will instantly fail.
- **Non-Repudiation:** Once Alice signs it, she cannot claim "I didn't send that." Since her public key successfully unlocks the signature, it proves her private key was used.

Every time your phone downloads an official app update, it verifies a digital signature from Apple or Google to ensure the code hasn't been injected with malware. Every time you connect securely to your bank, digital signatures are verifying that the website is legitimate.